# DR. WASIM AHMAD KHAN

**Machine Learning Engineer**

**Location:** Lahore, Pakistan | **Email:** drwasimahmadkhan@gmail.com | **Contact:** +92-322-4645667

## SUMMARY

I am an ambitious learner, software engineer, academician and researcher with extensive experience in the field of computer science. Currently, I am working as a Machine Learning Engineer at Techtics.ai, Lahore. Adept at leading teams to develop NLP-driven applications for edge devices using LLM technologies, while consistently achieving exceptional results. I was working as Associate Professor in Computer Science Department at NCBA&E Lahore and served as Head of Department Computer Science for more than three years. Furthermore, I bring practical experience to academia, having worked as a software engineer after graduation with a focus on server-side web development. My commitment to mentorship is evident through the supervision of student research projects and my active engagement in undergraduate policies discussions with HEC Lahore and the National Computing Education Accreditation Council Islamabad. I have leveraged this experience to facilitate meaningful partnerships, having initiated MoUs with multiple software houses to align academic programs with industry needs. Throughout my career, my research has focused on Applied Artificial Intelligence and Artificial General Intelligence, with particular emphasis on Cloud Robotics, ROS, and Cognitive Agents. I have extensive experience in programming languages such as Kotlin, Java, Python, PHP .Net, C#, and C++. I am always seeking new opportunities to learn and grow in this exciting and dynamic field.

### AREAS OF EXPERTISE

| | |
|---|---|
| ***Research Interests:*** | Artificial Intelligence, Cloud Robotics, Cognitive Agents, Social behaviours (Attention Regulation) of Robots, Machine Learning, Deep Learning, Machine Consciousness, Natural Language Processing, Conversational AI, Artificial General Intelligence, and Quantum Computing. |
| ***Technical Skills:*** | C++, Python, Java, C#, PHP, ASP.Net, JSP, MySQL, SQL Server, Power BI, Tableau, TensorFlow, TensorFlow Lite, PyTorch, Scikit-learn, Hugging Face Transformers, LLMs (fine-tuning & deployment), Vector Databases (FAISS, Chroma, Weaviate), LangChain, MLOps (MLflow, DVC), DevOps (Docker, GitHub Actions, CI/CD), Kubernetes, RESTful APIs, NLP, Computer Vision, iCub Simulator, YARP, ROS, Arduino, Raspberry Pi 3 B+. |
| ***Languages:*** | English, Urdu, Arabic, Punjabi |

## PROFESSIONAL EXPERIENCE

**Techtics.ai,** Lahore, Pakistan **13, Oct 2023 – Present**

*MACHINE LEARNING ENGINEER*

- Engineered a high-precision Computer Vision pipeline for jewelry retrieval, utilizing BiRefNet and YOLO for 99% accurate background removal and real-time ring matching.
- Architected an autonomous procurement engine using Gemini 2.5 Flash to parse unstructured PDFs into JSON, validated by a custom Structural Semantic WER framework.
- Developed an LLM-driven data integrity agent for autonomous schema detection and dynamic error correction, ensuring 98% compliance in large-scale CSV datasets.
- Developed a comprehensive Transformer learning hub featuring an interactive, end-to-end visualizer that decomposes the entire word generation process into granular, step-by-step mathematical calculations.
- Developing a cutting-edge healthcare application for Android edge devices, implementing offline speech-to-text and Large Language Model (LLM) techniques using VOSK, Whisper AI, Pytorch, and TensorFlow Lite. Achieved a remarkable reduction in processing time from 20-30 minutes to under 2 minutes for patient report diagnosis.
- Contributed to enhancing the reliability and efficiency of a live production application linked to the cloud, focusing on optimizing analysis and resolving speaker crosstalk detection issues, thereby ensuring a more robust and efficient user experience.

**INTI International University,** Nilai, Negeri Sembilan, Malaysia **06, May 2025 – Present**

*RESEARCH FELLOW — REMOTE*

- Conducting collaborative research in computer science, focusing on emerging technologies and interdisciplinary innovation.
- Engaged in funded projects and postgraduate supervision, while actively contributing to research seminars and academic development programs.

**NATIONAL COLLEGE OF BUSINESS ADMINISTRATION & ECONOMICS,** Lahore, Pakistan **Sep 2016–Oct 2023**

*ASSOCIATE PROFESSOR — DEPARTMENT OF COMPUTER SCIENCE* **07, Apr 2022 – 12, Oct 2023**
*HEAD OF DEPARTMENT COMPUTER SCIENCE* **15, Oct 2019 – 02, Mar 2023**
*ASSISTANT PROFESSOR* **15, Sep 2016 – 06, Apr 2022**

- I delivered engaging lectures, guided student research, and contributed to academic leadership and research in computer science, artificial intelligence, and robotics.
- Led and managed the Computer Science Department at NCBA&E (Private University), overseeing a team of 20+ faculty members. Focused on achieving excellence in academic activities and departmental development.
- I mentored students in courses such as Machine Learning, Web Engineering, Object-Oriented Programming, and Programming Fundamentals, aligning the curriculum with state-of-the-art industry standards.

**UNIVERSITY OF THE PUNJAB (PUCIT),** Lahore, Pakistan **Oct 2014 – Jul 2023**
**VISITING FACULTY — *DEPARTMENT OF COMPUTER SCIENCE***

- Instructed various courses using diverse teaching aids and engagement strategies to promote active learning and stimulate curiosity.
- Specialized in delivering lectures on Mobile Computing (Android development with JAVA), Enterprise Application Development (C#, ASP.Net), Theory of Automata, and Web Engineering (JAVA, JSP) to senior semesters.

**EICONIX INC,** Lahore, Pakistan **Jul 2011 – Sep 2016**
**SENIOR SOFTWARE ENGINEER** **Sep 2014 – Sep 2016**
**SOFTWARE ENGINEER** **Jul 2011 – Aug 2014**

- Managed high-priority web development projects, led a dynamic team in coordination, requirements derivation, and software design.
- Proficient in PHP (Custom PHP, Wordpress, and Custom CMS), and contributed to Android app development using JQuery Mobile.

## DESIGN PATENT

| No. | DESIGN PATENT(LINK WITH ACHIEVEMENTS) |
|---|---|
| 1 | **Wasim Ahmad Khan**, et. al. 2018. "Gyro-Hit (Virtual Reality Defence Training System-VRDTS)". Intellectual Property Organization Of Pakistan, Design No. '18926-D', filed September 25, 2017, and issued April 16, 2018 |

- 1st position in SoftExpo '17 Software competition, Annual National Competition at Punjab University College of Information Technology, Lahore.
- 1st position in SOFTEC '17 Software competition, Annual National Competition at FAST NUCES, Lahore.
- 1st position in NASCON '17 Software competition, Annual National Competition at FAST NUCES, Lahore.
- 1st position in GEEK WEEK '17 Software competition, Annual National Competition at FAST NUCES, Lahore.
- 1st position in ASCII '17 Software competition, Annual National Competition at International Islamic University, Islamabad.
- Regional Winner and National 2nd Runner up in Microsoft Imagine Cup '17, Islamabad.

## RESEARCH EXPERIENCE AND PUBLICATIONS

| No. | JOURNAL PAPERS (Accumulative Impact Factor: 34.296) |
|---|---|
| 26 | Abbas, S., Ahmed, F., **Khan, W. A.**, et al. (11, Jan. **2025**). **Intelligent skin disease prediction system using transfer learning and explainable artificial intelligence.** *Scientific Reports*, **ISSN 2045-2322,** *15*(1), 1746. [IF:3.8, Category: W, https://doi.org/10.1038/s41598-024-83966-4] |
| 25 | Ahmed, F., Abbas, S., Athar, A., Shahzad, T., **Khan, W. A.**, et al. (06, May **2024**). **Identification of kidney stones in KUB X-ray images using VGG16 empowered with explainable artificial intelligence.** *Scientific Reports*, **ISSN 2045-2322,** *14*(1), 6173. [IF:4.6, Category: W, https://doi.org/10.1038/s41598-024-60554-0] |

24 Alam, T. M., Shaukat, K., **Khan, W. A.**, et al. (31, Aug. **2022**). **An Efficient Deep Learning-Based Skin Cancer Classifier for an Imbalanced Dataset**. *Diagnostics*, **ISSN 2075-4418**, 12(9), 2115, 1-16. [IF:3.992, Category: X, https://doi.org/10.3390/diagnostics12092115]

23 **Khan, W. A.**, et al. (28, Jan. **2022**). **Trust identification through cognitive correlates with emphasizing attention in cloud robotics**. *Egyptian Informatics Journal,* **ISSN 1110-8665**, 23(2), 259-269. [IF:3.943, Category: W, https://doi.org/10.1016/j.eij.2022.01.003]

22 Khan, A. H., Abbas, S., Khan, M. A., Farooq, U., **Khan, W. A.**, et al. (21, Jan. **2022**). **Intelligent Model for Brain Tumor Identification Using Deep Learning**. *Applied Computational Intelligence and Soft Computing,* **ISSN 1687-9732**, 2022(Article ID 8104054), 1-10. [Category: X, https://doi.org/10.1155/2022/8104054]

21 Alam, T. M., Shaukat, K., Khelifi, A., **Khan, W. A.**, et al. (11, Oct. **2021**). **Disease Diagnosis System Using IoT Empowered with Fuzzy Inference System**. *CMC-Computers, Materials & Continua,* **ISSN 1546-2226**, 70(3), 5305-5319, USA. [IF:3.772, Category: W, https://doi.org/*10.32604/cmc.2022.020344*]

20 Khan, Q-T-A., Ghazal, M. T., Abbas, S., **Khan, W. A.**, et al. (11, Aug. **2021**). **Modeling Habit Patterns Using Conditional Reflexes in Agency**. *Intelligent Automation & Soft Computing,* **ISSN 1079-8587**, 30(2), 539-552. [IF:1.647, Category: X, https://doi.org/10.32604/iasc.2021.018888]

19 Daoud, M. S., Fatima, A., **Khan, W. A.**, et al. (07, Sep. **2021**). **Joint Channel and Multi-User Detection Empowered with Machine Learning**. *CMC-Computers, Materials & Continua,* **ISSN 1546-2226**, 70(1), 109–121, USA. [IF:3.772, Category: W, https://doi.org/10.32604/cmc.2022.019295]

18 Alam, T. M., Shaukat, K., Hameed, I. A., **Khan, W. A.**, et al. (15, May **2021**). **A novel framework for prognostic factors identification of malignant mesothelioma through association rule mining**. *Biomedical Signal Processing and Control*, **ISSN 1746-8094**, *68(Article ID 102726)*. [IF:3.88, Category: W, https://doi.org/10.1016/j.bspc.2021.102726]

17 Ahmad, F., Almuayqil, S. N., Humayun, M., Naseem, S., **Khan, W. A.**, et al. (28, Dec. **2020**). **Prediction of COVID-19 cases using machine learning for effective public health management**. *CMC-Computers, Materials and Continua*, **ISSN 1546-2226**, 66(3), 2265-2282, USA. [IF:4.89, Category: W, https://doi.org/10.32604/cmc.2021.013067]

16 Ali, N., Ahmed, A., Abbas, S., Ali, H., Fatima, A., and **Khan, W. A.** (14, Apr. **2020**). **Globalization: Information Technology and Economic Growth Nexus: Empirical Evidence from ARDL Model Approach to Co-integration and Causality Analysis.** *European Online Journal of Natural and Social Sciences*, **ISSN 1805-3602**, 9(2), 265-285. [Category: X, https://european-science.com/eojnss/article/view/5341]

15 **Khan, W. A.**, Abbas, S., Khan, M. A., et al. (09, Mar. **2020**). **Intelligent task planner for cloud robotics using level of attention empowered with fuzzy system.** *SN Applied Sciences*, **ISSN 2523-3971**, 2(4), 1-13. [Category: X, https://doi.org/10.1007/s42452-020-2312-4]

14 Butt, D. H., & **Khan, W. A.** (07, Jan. **2020**). **Theoretical Framework of Agent Assisted Treatment for ASD (Autism Spectrum Disorder).** *VFAST Transactions on Computer Sciences*, **ISSN 2308-8168**, 8(1), 36-47. [http://dx.doi.org/10.21015/vtcs.v15i2.515]

13 Abbas, S., Alyas, T., Athar, A., Khan, M. A., Fatima, A., **Khan, W. A.** (04, Jul. **2019**). **Cloud Services Ranking by measuring Multiple Parameters using AFIS**. *EAI Endorsed Transactions on Scalable Information Systems*, **ISSN 2032-9407**, 6(18:e4), 1-7, Belgium. [Category: X, http://dx.doi.org/10.4108/eai.13-7-2018.159354]

12 **Khan, W. A.**, et al. (13, Apr. **2018**). **Intelligent Virtual Security System using Attention Mechanism**. *EAI Endorsed Transactions on Scalable Information Systems*, **ISSN 2032-9407**, 5(16:e7), 1-6, Belgium [Category: X, https://doi.org/10.4108/eai.13-4-2018.154473]

11 **Khan, W. A.**, et al. (30, Jun. **2017**). **Identification of Associations between Cognitive Agents using Learning based System**. *Research Journal of Computer Science and Information Technology*, **ISSN 2519-7991**, 1(2), 62-78, LGU, Lahore, Pakistan. [Category: Y, http://lgurjcsit.lgu.edu.pk/index.php/lgurjcsit/article/view/29]

10 Athar, A., Ahmed, K., Saeed, Y., **Khan, W. A.**, et al. (01, Jan. **2017**). **Modeling Emotional Abstraction of sensory perceptual content for cognitive agents using fuzzy Inference.** *International Journal of Computer Science and Information Security*, **ISSN 1947-5500,** 15(1), 491-496, Pittsburgh, PA, USA. [Category: X, https://sites.google.com/site/ijcsis/all-volumes-issues/vol-15-no-1-jan-2017]

9 **Khan, W. A.**, et al. (20, Dec. **2016**). **Integrated Approach of Human-Robot Interaction on Basis of Top-Down and Bottom-Up Attention Approach Using Visual and Verbal Affective Intent in Agents**, *International Journal of Soft Computing and Artificial Intelligence (IJSCAI)*, pp. 26-31, Volume-4, Issue-2.

8 Abbas, S., Bilal, M., **Khan, W. A.**, et al. (11, Nov. **2016**). **Intelligent Agent Navigation using Bluetooth based Adhoc Communication.** *International Journal of Computer Science and Information Security,* **ISSN 1947-5500,** 14(11), 646-653, Pittsburgh, PA, USA. [Category: X, https://sites.google.com/site/ijcsis/all-volumes-issues/vol-14-no-11-nov-2016]

7 Khan, M. W., **Khan, W. A.**, et al. (11, Nov. **2016**). **Strategy Development through Emotional Manipulation in Agents using Q-Learning.** *International Journal of Computer Science and Information Security,* **ISSN 1947-5500**, 14(11), 395-397, Pittsburgh, PA, USA. [Category: X, https://sites.google.com/site/ijcsis/all-volumes-issues/vol-14-no-11-nov-2016]

6 Tooba, **Khan, W. A.**, et al. (11, Jul. **2016**). **Cognitive Architecture to Generate Motivational Feelings: A Way to Improve Visual Learning in Robots.** *Computer Engineering and Intelligent Systems,* **ISSN 2222-1719**, 7(6), 37-40**.** [https://iiste.org/Journals/index.php/CEIS/article/view/30938]

5 Naseem, S., Abid, M. I., Butt, A. A., Ahmad, F., **Khan, W. A.**, et al. (1, Mar. **2016**). **Graphical User Interface of Fuzzy Logic based Distributed Flood Adaptive Control System for Agricultural Land and Power Generation in Punjab, Pakistan.** *Pakistan Journal of Science,* **ISSN 0030-9877**, 68(1), 31-41. [Category: X, http://paas-pk.com/SubPages/Journal/Dashboard.aspx?fcso=2&menuid=1004&JID=298]

4 Ahmad, F., Naseem, S., Alyas, T., Tabassum, N., **Khan. W. A.**, et al. (14, Sep. **2015**). **Forecasting of Intellectual Capital by Measuring Innovation using Adaptive Neuro-Fuzzy Inference System.** *International Review of Applied Sciences,* **ISSN 2611-667X**, 2(1), 1-13**.** [https://ideas.repec.org/a/aoj/inroas/2015p1-13.html]

3 **Khan, W. A.**, et al. (15, Jun. **2015**). **Cognitive System for Objects Serving in Industry.** *Computer Engineering and Intelligent Systems,* **ISSN 2222-1719**, 6(1), 26-29. [https://iiste.org/Journals/index.php/CEIS/article/view/190 55/19300]

2 **Khan, W. A.**, et al. (08, May **2015**). **Hibernation Mechanism in Smartphone Mobile Operating Systems.** *Journal of Energy Technologies and Policy,* **ISSN 2224-3232**, 5(1), 1-4. [https://www.iiste.org/Journals/index.php/JETP/article/view/19384]

1 Hameed, U., Naseem, S., Ahamd, F., Alyas, T., and **Khan, W. A.** (22, Dec. **2014**). **Intrusion Detection and Prevention in Cloud Computing using Genetic Algorithm.** *International Journal of Scientific & Engineering Research,* **ISSN 2229-5518**, 5(12), 1271-1275.

[https://www.ijser.org/onlineResearchPaperViewer.aspx?Intrusion-Detection-and-Prevention-in-Cloud-Computing-using-Genetic-Algorithm.pdf]

| No. | CONFERENCE PAPERS / PRESENTATIONS |
|---|---|

17 **Khan, W. A.** (23rd November, **2023**). **Current State of Natural Language Processing (NLP) and its Evolution Towards Artificial General Intelligence (AGI)**. Conference on Cyber Secure Pakistan Vision 2030, IAC, Lahore, **Pakistan**.

16 **Khan, W. A.**, et al. (2nd to 4th September, **2021**). **A Review on Security Challenges in Internet of Things (IoT)**. The 26th International Conference on Automation and Computing (ICAC'21), IEEE, Portsmouth, **UK**. [http://dx.doi.org/10.23919/ICAC50006.2021.9594183]

15 **Khan, W. A.**, et al. **Early Diagnosis of Dengue Symptoms Using Neuro**. *The 6th International Conference on Next Generation Computing (17th to 19th December **2020**), at* Shilla Stay Haeundae, Busan, **Korea**. [http://www.kingpc.or.kr/wp/?pageid=2&page_id=3350&uid=329&mod=document]

14 **Khan, W. A. Cloud based Security System using Cognition, and Attention Mechanism.** *3rd National Conference on CS & IT (23rd & 24th April **2019**),* at Lahore Garrison University Lahore, **Pakistan**.

13 **Khan, W. A. IoT will become part of Cognitive Computing.** *1st International Conference on Science, Technology and Innovation (2nd to 4th April **2019**),* at Kinnaird College for Women Lahore, **Pakistan**.

12 **Khan, W. A. Cloud Based Knowledge Sharing in Robotics for Attention Mechanism.** *1st International Conference on Science, Technology and Innovation (2nd to 4th April **2019**),* at Kinnaird College for Women Lahore, **Pakistan**.

11 **Khan, W. A. Knowledge Processing in Cloud Robotics for Attention Mechanism.** *1st International Conference on Computer Science and Allied Technologies (19th & 20th December **2018**),* at Lahore Garrison University Lahore, **Pakistan**.

10 **Khan, W. A.**, et al. **Trust Development in Social Cognitive Agents.** *1st International Conference on Action Research in Multidisciplinary Perspectives, (19th & 20th March, **2018**),* at University of Management & Technology Lahore, **Pakistan**.

9 **Khan, W. A.**, et al. **Lexicon Based Sentiment Classification for Web-based Urdu Text.** *16th International Conference on Statistical Sciences, (5th to 8th March, **2018**),* at Islamia College, Peshawar, **Pakistan**.

8 **Khan, W. A.**, et al. **Navigation Mesh Rendering, Object, Verbal and Human Recognition Framework for Robot Operating System.** *16th International Conference on Statistical Sciences, (5th to 8th March, **2018**),* at Islamia College, Peshawar, **Pakistan**.

7 **Khan, W. A. Attention Mechanism in Cloud Robotics.** *2nd National Conference on Information Technology, (28th & 29th March, **2018**),* at Lahore Garrison University Lahore, **Pakistan**.

6 **Khan, W. A. Surveillance Security System using Cognition, and Attention Mechanism.** *EAI International Conference on Future Intelligent Vehicular Technologies, (17th to 19th October, **2017**),* at Islamabad, **Pakistan**.

5 **Khan, W. A.**, et al. **Effective Load Balancing in Cloud Robotics using Rapuyta.** *1st National Conference on Information Technology, (3rd & 4th May, **2017**),* at Lahore Garrison University Lahore, **Pakistan**.

4 **Khan, W. A.**, et al. **Ethical Machine Learning through Sensors based Neural Network System and Intellectual Productivity Generation with reasonable Decision Making.** *1st National Conference on Information Technology, (3rd & 4th May, **2017**),* at Lahore Garrison University Lahore, **Pakistan**.

3 **Khan, W. A.**, et al. **Anti-Hacking Library for Developers using Adaptive Learning Techniques.** *14th International Conference on Statistical Sciences, (12th to 15th December, **2016**)* at Multan, **Pakistan**.

2 **Khan, W. A.**, et al. **Self-Awareness Mechanism for Top Down Attention using Fuzzy Logic in Cognitive Agents.** *1st International Conference on Science, Technology and Innovation Policy and Management, (16th & 17th November, **2016**),* at Mövenpick Hotel Karachi, Sindh **Pakistan**.

1 **Khan, W. A.**, et al. **A Conceptual Framework for Load Balancing in Working Memory of Social Robots.** *1st International Conference on Science, Technology and Innovation Policy and Management, (16th & 17th November, **2016**),* at Mövenpick Hotel Karachi, Sindh **Pakistan**.

## EDUCATION & OTHER CREDENTIALS

DEPARTMENT OF COMPUTER SCIENCE – NCBA&E, **Ph.D. in Computer Science 2020**
**Pakistan Country Directory Number (HEC): 20768**
***Dissertation: "Knowledge Sharing for Task Planning in Cloud Robotics."***

DEPARTMENT OF COMPUTER SCIENCE – NCBA&E, **M.Phil. in Computer Science, 2015**
***Dissertation: "Attention Regulatory System for Modulating Multiparty Interaction in Cognitive Agents."***

PUNJAB UNIVERSITY COLLEGE OF INFORMATION TECHNOLOGY, **M.Sc. in Information Technology, 2011**

### PROFESSIONAL AFFILIATIONS

Member of International Association of Engineers as Computer Scientists, located at Hong Kong
Member of the society for Interdisciplinary Research at NCBA&E